



A Fourier feature-embedded physics-informed neural network framework to investigate coupled heat and mass transfer characteristics of plant tissues during drying

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ABSTRACT

This paper introduces a new computational framework for analysing heat and mass transfer during the drying of plant tissues, using Fourier Feature Embedded Physics-Informed Neural Networks (FFE-PINN). The proposed FFE-PINN framework enables direct communication between two physics-informed neural network-based models: PINN-MT for mass transfer and PINN-HT for heat transfer, which are trained simultaneously to investigate heat and mass transfer characteristics. The novelty of this study is the integration of Fourier Feature Embedding (FFE) into the PINN framework to examine coupled heat and mass transfer during drying to significantly improve the accuracy and robustness of drying-kinetics-based predictions. The developed model demonstrates strong alignment with experimental data on moisture content variation and numerical results from Finite Element Analysis (FEA), with maximum deviations of 8.73% for moisture concentration and 4.51% for temperature predictions. The findings indicated that these differences are primarily due the distinct derivation techniques utilised in PINN and FEA, rather than any limitations of the proposed framework. Importantly, this study marks a significant milestone as the first to apply a PINN-based approach to analyse coupled heat and mass transfer in dried plant tissues over an extended 60-minute drying period, without relying on transfer learning, due to FFE introduction. The proposed FFE-PINN framework emerges as a promising computational tool, offering a physics-consistent approach to predict complex and nonlinear heat and mass transfer phenomena associated not only with drying, but further beyond.

1. Introduction

Fruits and vegetables, characterised by their heterogeneous, porous, and moisturised nature, are highly perishable due to microbial spoilage. Drying is a widely employed method for extending their shelf life by reducing moisture content, thereby inhibiting the growth of lactic acid bacteria, yeasts, and moulds [1]. This process not only extends the usability of these foods but also improves their transportability and storage efficiency due to reduced weight and volume. When performed under controlled conditions, drying preserves a significant portion of the nutritional content, such as vitamins, minerals, and fibre, contributing to its global relevance in reducing food waste and enhancing food

availability during off-seasons or in regions where cultivation is impractical [2].

However, drying remains a very complex process steered by unsteady heat and mass transfer phenomena that result structural and microstructural changes in plant-based foods. The inherent challenges of this energy intensive process are substantial, as it accounts for around 15 % of all industrial energy consumption [3]. Addressing these complexities requires a thorough understanding of the underlying heat and mass transfer mechanisms, spanning from microscale interactions and behaviours within the porous structure to macroscale industrial processes. While experimental investigations provide valuable insights, their scope is often limited by difficulties in measuring transient and spatially

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varying parameters such as temperature, moisture content, and shrinkage, especially within the internal structure of the material [4]. In response, physics-based mathematical models have been developed to simulate these intricate dynamics [5]. These models typically employ partial differential equations (PDEs) to describe relevant physical phenomena, solved under defined boundary (BC) and initial conditions (IC). Analytical solutions for such physics based mathematical models are usually feasible for one-dimensional (1D) problems with simple geometries and physical properties. However, the drying of plant-based foods involves simultaneous transient heat, mass, and momentum transfer in irregular and dynamically evolving microstructures with their mechanical deformations, necessitating multiphysics models that integrate coupled phenomena and nonlinear material property variations [6].

For complex drying applications, numerical computational approaches such as Finite Element Method (FEM), Finite Difference Method (FDM), and meshfree approaches - including Smoothed Particle Hydrodynamics (SPH) and Discrete Element Method (DEM), are widely utilised [7]. These methods have yielded significant insights into drying kinetics and microstructural behaviour. For example, in early mesh-based FEA analyses, a two dimensional (2D) virtual microstructure tissue model has been developed to study dehydration characteristics and mechanical deformation in pear fruit tissues by combining mass transfer and momentum physics [8]. A similar investigation has later been conducted for apple tissues, incorporating viscoelastic properties in the deformation model [9]. More recently, researchers have attempted to simplify the model domain by employing a representative element approach and an upscaling methodology. This approach couples microscale to macroscale behaviours through permeability calculations, which have been used to study the dehydration of apple tissues [10]. However, these methods predominantly focus on dehydration at or near room temperature.

Within the drying context, some numerical models initially excluded temperature variations but incorporated multiphase behaviour through meshfree, particle-based methods [4,11,12]. Unlike mesh-based approaches, the domain of meshfree models is not constrained by permanent node-to-node connections. Instead, the domain is represented by particles that carry both geometric and material properties. These particles interact dynamically with neighbouring particles during simulation. Such flexibility has enabled new insights into drying kinetics by capturing large deformations and coupling solid dynamics with fluid mechanics [7,13]. Meshfree studies have further expanded to include the analysis of three-dimensional (3D) plant cell drying characteristics [14,15], and to examine porosity and case-hardening effects in plant tissues during drying [16,17]. However, these meshfree modelling approaches did not consider heat transfer effects. In parallel, researchers developed coupled heat and mass transfer models to investigate moisture transport properties influenced by temperature variations [18]. Perussello, Kumar, de Castilhos and Karim [19] developed a FEM-based heat and mass transfer model to study temperature and moisture variations along with sucralose content in plant tissues. Utilising a free mesh generator and code-based simulations, another study investigated the impact of vibrating hot airflow on heat and mass transfer. The results demonstrated that drying could be accelerated using an oscillating convective drying flow [6]. In another FEM study, the heat and mass transfer process in a single peanut has been investigated using mass and energy balance principles [20]. The results have been validated against thin layer drying experiments, revealing that the FEM model accurately predicted moisture content and provided insights into heat and mass transfer mechanisms during drying. Multiscale models within FEM-based analyses have also been applied to enhance the predictive capabilities of heat and mass transfer at various temperatures, incorporating generalized diffusivity and microscale cell rupture effects [21, 22]. While these models have been developed coupling heat and mass transfer physics models, Tuly, Joardder, Welsh and Karim [23] recently developed a FEM model to investigate drying kinetics of jackfruit by coupling heat and mass transfer along with shrinkage model defined

based on ratio of the transient volume to initial volume.

Overall, numerical simulations employing both mesh-based and meshfree methods offer valuable insights into the behaviour of plant-based foods during drying. However, traditional computational models are constrained by method-specific limitations. Mesh-based methods often struggle with challenges such as mesh generation, refinement, and distortion, particularly in complex geometries or large deformations [7]. On the other hand, meshfree methods face difficulties with numerical integration, consistency, and high computational costs, especially in three-dimensional simulations [24]. Moreover, no heat transfer effects have yet been considered within mesh-free particle-based approaches within food drying context. These limitations have restricted their broader applicability in certain scenarios. In response, there has been a growing interest in data-driven modelling approaches, which offer the potential to overcome these challenges by leveraging advanced techniques such as machine learning and physics-informed neural networks (PINNs) [25,26]. Data-driven approaches do not compute physical phenomena directly but instead predict variations or trends based on historical data. In the context of food drying, machine learning models have been developed to predict heat and mass transfer variations [27] and to optimize drying kinetics [25,28]. These models can also be utilised for real-time drying process control by leveraging real-time monitoring sensors, such as computer vision, making artificial intelligence-based approaches particularly valuable in the food drying industry [25,28]. A recent study addressed challenges associated with traditional model predictive control in drying by introducing a deep learning-based predictive model for industrial-scale paddy drying. This study established a fast prediction framework for a counter-flow drying system using small-sample modelling techniques combined with generalized optimization algorithms and support vector mechanisms [29]. Machine learning models generally offer significantly faster prediction speeds, even under previously unseen conditions. However, while sufficient data can often be collected for real-time drying process control, a realistic limitation exists in obtaining adequate data for training machine learning models to predict the complex drying kinetics within plant-based foods. Consequently, the investigation of these variations remains an unaddressed challenge with purely data-driven machine learning approaches [25].

Recently, PINN models have emerged as a promising solution to overcome the limitations of both traditional computational approaches and data-driven machine learning methods [30]. PINNs are a specialised category of artificial neural networks that can integrate physics principles with relevant data. This unique capability enables them to extract information from both datasets and physics-based equations or conditions. As a result, PINNs offer significantly improved accuracy for problems involving ill-posed, noisy, or insufficient data, thanks to their additional support on embedded physical laws during training [24]. Moreover, unknown theoretical conditions, process parameters, product properties, and transient material behaviours can be uncovered through the synergistic use of physics-based principles and available data. To achieve these advantages, the training process for PINNs must be guided by both datasets and related physics constraints [31]. As an alternative to traditional computational methods, PINNs can also solve physics-based models without labelled data when the loss function is defined exclusively by physical conditions [24]. A key feature of PINNs is the computation of PDE operators using automatic differentiation (AD) within the neural network framework [32]. This approach contrasts with the numerical differentiation employed in traditional computational techniques. Furthermore, the mesh-free domain of PINNs provides enhanced flexibility for modelling dynamic systems, especially those with complex geometries, large deformation, and nonlinear behaviours [33–35].

In heat transfer field, PINN models have been successfully implemented recently. Jalili and Mahmoudi [36] have developed a PINN framework to analyse two-phase film boiling, where the network was trained on CFD-generated data and subsequently adapted through

transfer learning to predict interface evolution and Nusselt number at different Jakob numbers. Similarly, Jalili, Jang, Jadidi, Giustini, Keshmiri and Mahmoudi [37] employed a data-driven PINN to model two-phase flows with heat transfer, training on high-fidelity CFD simulations to capture bubble rise, interface dynamics and the thermal wake near heated walls, and demonstrating accurate forward and inverse predictions across varying fluid properties. These works establish PINN as a powerful surrogate for multiphase convection and boiling problems relying on traditional simulation data and transfer learning. By contrast, this study develops a dual-network FFE-PINN framework that directly solves coupled heat and mass transfer PDEs in plant tissue drying using only physics-based residuals, without any data, pre-training or transfer learning.

Authors of this article begun exploring the potential of novel PINN computational approaches in the field of food drying as an alternative to existing computational and simulation models [13,24,38]. These efforts aim to address the current challenges within the drying context. Initially, a PINN surrogate model was developed to investigate mass transfer and shrinkage variations in plant cells [39]. This study demonstrated the strong predictive capabilities of PINN models in handling unknown process and product conditions and highlighted their ability to couple physics-based equations with inherent machine learning surrogate modelling techniques. Moreover, the study emphasized the seamless integration of real-world data with generated data from physics equations, enabling predictions at a more reliable and accurate level. Subsequently, a PINN model was developed to evaluate the efficacy of AD in solving a Fickian diffusion mass transfer physics model for single-cell domains with no label data [32]. This research was further extended to simultaneously solve coupled mass transfer and shrinkage models within a single-cell domain [34]. Domain decomposition, combined with the neural network's integration capabilities, was utilized to introduce heterogeneous boundary conditions and nonuniform material properties to the single cell drying process. While these prior studies have primarily focused on single-cell domains, the current investigation introduces a novel PINN-based computational framework to investigate temperature-driven moisture transport variations in plant tissues during drying. As a novel computational approach, PINN has not yet been capable of solving transient variations of heat and mass transfer for a realistic drying time. While Manavi, Fattahi and Becker [40] developed a parameterised PINN model for heat and mass transfer, their work was limited to a 1D domain with 5-mins drying duration and lacked comprehensive comparison of spatial variations. The reason for such limitations is due to lack of learning capabilities of current PINNs. Recent attempts introducing several features to PINNs can be highlighted trying to overcome such limitations. Physics-informed kernel function neural networks (PIKFNNs) embedding physical information of governing equations into the activation functions rather than the loss function, enabling the training of neural networks using only boundary and initial data [41]. Another improvements include transfer learning with pre-training to improve the convergence and accuracy of standard PINNs for solving strongly nonlinear or high-frequency time-evolution PDEs by dividing the time domain into smaller subdomains [42–44]. Similarly, to avoid lack of training in traditional PINNs, in this study we introduced Fourier feature-embedded (FFE) layer to the feed forward neural network enhancing PINN's learning capabilities. Due to this novel development, the FFE-PINN framework is capable of solving heat and mass transfer with coupled effects for 60 mins drying time, being the first to achieve such capabilities within food drying context.

This paper is structured as follows: Section 2 presents the physics-based models employed in the study and the domain considerations. Section 3 details the development of the PINN model and its associated parameters. In Section 4, the capabilities of the proposed PINN framework are evaluated, with results from case studies, and corresponding discussions are presented in Section 5. Finally, the paper concludes with remarks presented in Section 6.

2. Physics-based model development for heat and mass transfer of drying tissue

This section outlines the geometry of the plant food sample considered for the simulation, and the physics principles incorporated into the PINN-based computational framework including the heat and mass transfer coupling approach.

2.1. 2D tissue geometry

The heat and mass transfer PINN models were developed for a small two-dimensional (2D) tissue domain, assuming symmetric boundaries from bottom and left edges, as shown in Fig. 1.

The 40 mm × 40 mm full-size tissue domain is simplified to a 20 mm × 20 mm quarter representation, applying symmetric conditions along the left and bottom boundaries while allowing flux to occur solely through the top and right boundaries. Such a quarter model with symmetric boundary conditions let to analyse just 1/4 of a symmetric structure while getting results identical to a full model, saving significant computational resources as long as both the geometry and loading conditions are symmetric.

2.2. Mass transfer model

During the drying of food products such as fruits and vegetables, the moisture content decreases over time as moisture evaporates from the surface. As moisture is lost from the boundaries, the moisture concentration within the food material changes, influenced by the rate at which moisture is transported within the material, a process known as diffusion. Fick's Law of Diffusion provides a fundamental description of how moisture moves from regions of high concentration (the interior of the material) to regions of low concentration (the surface) [45]. Fick's Law is considered as mass transfer model in this study to investigate moisture transport, and it can be mathematically expressed as [24]:

$$\frac{\partial c}{\partial t} + \nabla \cdot (-D \nabla c + uc) = S \quad (1)$$

where c represents the instantaneous moisture concentration in mol/m^3 , t is the drying time in seconds, D is the diffusivity in m^2/s , u is the convective flow velocity, and S is the source or sink term for moisture transport.

Neglecting the convective flow and source or sink terms, the equation can be simplified and applied to the 2D tissue model for this study as follows:

$$\frac{\partial c}{\partial t} = \nabla D \nabla c, \quad (2)$$

where ∇ represents the spatial derivative of a scalar field. As this work focus on 2D space as described in Fig. 1, the spatial coordinates are x and y in meters.

In the mass transfer model, the boundary conditions represent the interactions between the plant tissue and the surrounding drying environment during the drying process. The Neumann boundary condition is utilised to describe the boundary conditions, where moisture flux is proportional to the difference in concentration between the boundary of the material and the surrounding air and can be defined as [24]:

$$D \nabla c|_{un} = h_m (c_{bc} - c_{air}), \quad (3)$$

where h_m represents the mass transfer coefficient in the boundary in m/s and un denotes the direction of the mass transfer normal to the boundary. c_{bc} and c_{air} are moisture concentrations at boundary of the tissue where mass flux occurs and moisture concentration of air, respectively.

Symmetric boundary conditions can be defined as,

$$D \nabla c|_{un} = 0 \quad (4)$$

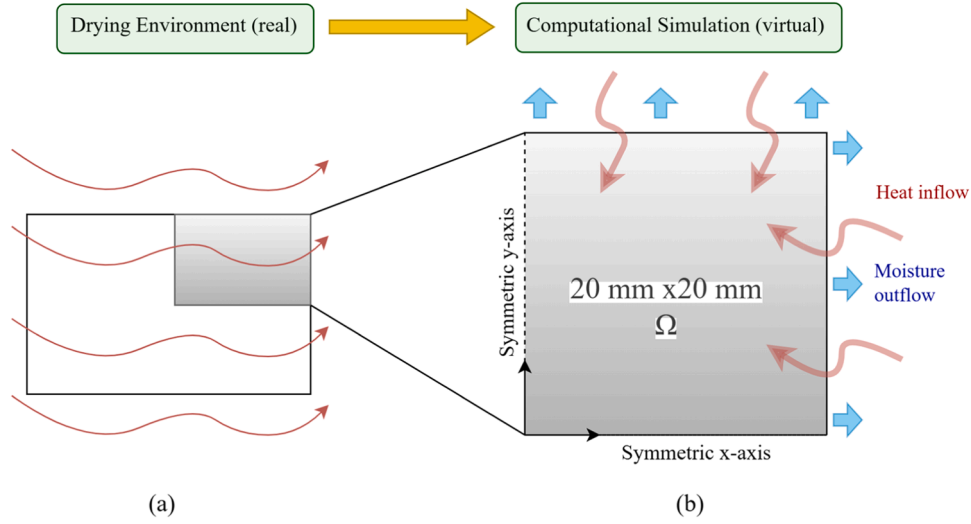


Fig. 1. Spatial domain representation of the drying tissue (a) Drying tissue sample in the drying environment (size 40 mm x 40 mm) and (b) A 20 mm × 20 mm domain with symmetric boundaries along the bottom and left edges.

Initial condition for the mass transfer model is the fresh moisture concentration of the plant tissue. It is a function of the initial tissue wet basis moisture content ($M_{wb,i}$) in kg/kg, the initial material density (ρ) in kg/m³, and the molar mass of water (M_w) in kg/mol [24]. Based on this, Dirichlet initial condition can be expressed as:

$$c_0|_{t=0} = \frac{M_{wb,i} \cdot \rho_i}{M_w} \quad (5)$$

2.3. Heat transfer model

The heat transfer model can be defined by considering the energy balance and applying Fourier's Law of heat conduction [46]. In the context of drying, heat is transferred from the surrounding dryer air or external source to the surface of the food material, where it subsequently facilitates the evaporation of moisture. As heat flows into the material, it raises the temperature, affecting the rate of moisture loss. Fourier's Law is used in this study to describe the heat conduction process within the tissue and it can be mathematically expressed as [47]:

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p \mathbf{u} \cdot \nabla T = \nabla \cdot (k \nabla T) + Q_e, \quad (6)$$

where ρ is the density of the material in kg/m³, C_p the specific heat of the material in J/(kg.K), T is the instantaneous temperature in Kelvin, k is the thermal conductivity of the material in W/(m.K), and Q_e the internal rate of heat generation.

By neglecting convective heat flow and internal heat generation, the above equation can be simplified and applied to a 2D tissue model as follows:

$$\rho C_p \frac{\partial T}{\partial t} = \nabla k \nabla T \quad (7)$$

The specific heat and thermal conductivity of the drying food material can be defined as transient properties, varying as a function of the instantaneous wet basis moisture content (M_{wb}) of the materials, as expressed below [21].

$$\begin{aligned} c_p &= 1000(1.4 + 3.22 M_{wb}) \\ k &= 0.148 + 0.49 M_{wb} \end{aligned} \quad (8)$$

In the heat transfer model, the boundary condition represents the interactions between the plant tissue boundaries and the surrounding drying environment during the drying process. The Neumann boundary condition is utilised to describe the heat flux at the boundaries, where

the rate of heat transfer within the boundaries of the tissue is proportional to the temperature difference between the material's surface and the surrounding air, as well as the heat losses due to the evaporation of moisture. This can be defined as:

$$k \nabla T|_{\text{um}} = h_T(T_{bc} - T_{air}) + h_m(c_{bc} - c_{air})\lambda M_w, \quad (9)$$

where h_T is the heat transfer coefficient in the boundary in W/(m².K), λ is the latent heat of evaporation in J/kg and M_w is the molar mass of the water in kg/mol. T_{bc} and T_{air} are temperature at boundary of the tissue where heat flux occurs and air temperature.

Symmetric boundary conditions can be defined as,

$$k \nabla T|_{\text{um}} = 0 \quad (10)$$

The Dirichlet initial condition can be defined based on the initial drying temperature for the heat transfer model.

2.4. Heat and mass transfer coupling

Fick's diffusion-based mass transfer model predicts the moisture concentration within tissue, which is influenced by the drying temperature. The moisture content within the tissue is related to the equilibrium vapor pressure (P_{eq}) which is a function of temperature (T) in Kelvin, the saturated vapor pressure of water (P_{sat}) in Pascals and the dry basis moisture content (M_{db}) in kg/kg. The equilibrium vapor pressure (P_{eq}) for apple in Pascals can be defined as [21]:

$$\begin{aligned} P_{v,eq} &= P_{v,sat}(T) \cdot \exp\left(\right. \\ &\quad \left. - 0.182 M_{db}^{0.696} + 0.232 e^{-43.949 M_{db}} M_{db}^{0.0411} \cdot \ln[P_{v,sat}(T)] \right) \end{aligned} \quad (11)$$

The saturated vapor pressure can be defined as:

$$\begin{aligned} P_{v,sat} &= \exp \left[-\frac{5800.2206}{T} + 1.3915 - 0.0486T + 0.4176 \cdot 10^{-4} T^2 \right. \\ &\quad \left. - 0.1445 \cdot 10^{-7} T^3 + 6.546 \ln(T) \right] \end{aligned} \quad (12)$$

The moisture concentration within the plant tissue in Eq. (3) was calculated based on the equilibrium vapor pressure ($P_{v,eq}$) in Eq. (11) and from ideal gas law as below:

$$c = \frac{P_{v,eq}}{RT} \quad (13)$$

Conversely, the heat transfer boundary condition is influenced by variations in moisture concentration due to evaporation, as shown in Eq.

(9). Further information is discussed in detail in Section 3 with the PINN architecture.

3. Physics informed neural network framework development for heat and mass transfer of drying tissue

This section presents the development of the PINN framework, which primarily comprises PINN-MT for predicting moisture concentration influenced by temperature variations and PINN-HT for predicting temperature distribution affected by moisture loss. The model incorporates the mutual interactions between moisture concentration and temperature distribution, ensuring updates are made during each training iteration to accurately capture these coupling effects.

3.1. PINN for predicting mass transfer (PINN-MT) in plant tissue

A fully connected feed-forward neural network is utilised including FFE and the training performed based on residual losses derived from the governing equation, boundary conditions, and initial conditions. These losses are quantified as mean squared errors.

The mean squared residual loss associated with the mass transfer governing equation (Eq. (2)) can be defined as:

$$MSE_{Eqn,MT} = \frac{1}{n_{\Omega}} \sum_{i=1}^{n_{\Omega}} \left(\frac{\partial c}{\partial t} - \nabla D \nabla c \right) \quad (14)$$

Similarly, the mean squared residual losses for the boundaries with mass flux and symmetric conditions can be derived from Eq. (3) and Eq. (4) respectively, as follows:

$$MSE_{BC,MT} = \frac{1}{n_{\bar{\Omega}}} \sum_{i=1}^{n_{\bar{\Omega}}} (D \nabla c|_{\text{un}} - h_m(c_{bc} - c_{air})) \quad (15)$$

$$MSE_{BC-S,MT} = \frac{1}{n_{\hat{\Omega}}} \sum_{i=1}^{n_{\hat{\Omega}}} (D \nabla c|_{\text{un}}) \quad (16)$$

The mean squared residual loss for the IC can be derived from Eq. (5) as follows:

$$MSE_{IC,MT} = \frac{1}{n_{\bar{\omega}}} \sum_{i=1}^{n_{\bar{\omega}}} \left[(c_{t=0}) - \left(\frac{M_{wb,i} \cdot \rho_i}{M_w} \right) \right] \quad (17)$$

Here, Ω represents the domain, $\bar{\Omega}$ denotes the boundary domain allowing for mass flux, $\hat{\Omega}$ represents the symmetric boundary domain and $\bar{\omega}$ corresponds to the initial condition domain. The subscripts *Eqn* and *BC-S* refers to the governing equation and symmetric boundary conditions, respectively.

3.2. PINN for predicting heat transfer (PINN-HT) in plant tissue

A development of the loss functions for PINN-HT model is discussed in this section. The mean squared residual losses corresponding to the heat transfer governing equation (Eq. (7)) can be expressed as:

$$MSE_{Eqn,HT} = \frac{1}{n_{\Omega}} \sum_{i=1}^{n_{\Omega}} \left(\rho C_p \frac{\partial T}{\partial t} - \nabla k \nabla T \right) \quad (18)$$

Similarly, the mean squared residual losses of BCs and IC related to the heat transfer model can be derived from Eq. (9) and Eq. (10) respectively, as:

$$MSE_{BC,HT} = \frac{1}{n_{\bar{\Omega}}} \sum_{i=1}^{n_{\bar{\Omega}}} (k \nabla T|_{\text{un}} - (h_T(T_{bc} - T_{air}) + h_m(c_{bc} - c_{air}) \lambda M_w)) \quad (19)$$

$$MSE_{BC-S,HT} = \frac{1}{n_{\hat{\Omega}}} \sum_{i=1}^{n_{\hat{\Omega}}} (D \nabla T|_{\text{un}}) \quad (20)$$

The mean squared residual loss for IC of the heat transfer can be formulated based on the initial drying temperature (T_0) as:

$$MSE_{IC,HT} = \frac{1}{n_{\bar{\omega}}} \sum_{i=1}^{n_{\bar{\omega}}} [(T_{t=0}) - T_0] \quad (21)$$

3.3. PINN-MT (mass transfer) and PINN-HT (heat transfer) network architecture with coupling

The coupled PINN complete architecture, along with the flow of information between the coupled networks, is illustrated in Fig. 2, highlighting the simultaneous training process guided by domain-specific physics and the mutual updating of parameters.

Both the PINN-MT and PINN-HT models are trained simultaneously within a single framework, with the models exchanging results throughout the training process. The moisture concentration is dependent on the instantaneous temperature, as described in Eq. (13). Therefore, the instantaneous temperature predicted by PINN-HT is used to calculate moisture during the training process. Likewise, the evaporation rate at the boundary, influenced by the temperature values, is incorporated into the boundary conditions of PINN-HT. This evaporation rate is derived from the moisture concentration at the boundary, which is calculated by PINN-MT and shared with PINN-HT for further updating. All MSE terms in the loss equation shown in Fig. 2 are equally weighted, as the Fourier Feature Embedding (FFE) automatically balances gradient contributions across physical residuals, ICs, and BCs without requiring manual tuning.

The monolithic coupling approach (one optimizer, shared collocation points) similar to the Deep Energy Method (DEM) in solid mechanics [48], is adopted so that PINN-MT and PINN-HT are updated simultaneously with a single optimisation loop. The coupling mechanism and real-time updates of temperature-dependent moisture transfer and moisture-dependent heat transfer are summarised in Algorithm below.

This optimisation loop is run till the loss converged or till max number of epochs reached.

3.4. Fourier features embedding (FFE) to PINN network

The NN architecture is a critical element in any PINN computational model, as it determines the PINN's capacity to learn and respect the complex physical relationships coupled into the model and its prediction accuracy. Typically, an NN consists of multiple layers, each containing several neurons interconnected by weighted and biased connections. The selection of the number of layers and neurons in each layer depends on the complexity of the problem and the need to capture non-linear interactions within the data. The activation function plays a pivotal role in introducing non-linearity into the model, which is essential for capturing the underlying physics-based parameters [24,49,50]. Therefore, the choice of the activation function is a critical decision. In this study, both the PINN models were implemented using the cosine activation function [51] and the Adam optimizer within the widely used open-source machine learning library, PyTorch [52]. Moreover, FFE was integrated into the PINN models in this study to improve the learning accuracy as discussed below.

FFE serves as a powerful input transformation technique that can significantly enhance NN's learning accuracy of high-frequency functions within the low-dimensional domain. Multilayer perceptrons (MLPs) are known to encounter challenges in learning high-frequency functions due to their inherent bias toward the representation of low-frequency functions, a phenomenon commonly referred to as 'spectral bias.' According to neural tangent kernel (NTK) theory, standard MLPs are associated with kernels that exhibit a rapid frequency falloff, thereby limiting their capacity to accurately represent high-frequency features [53,54]. This preprocessing step involves transforming the input data with Fourier feature mapping, which enables the tuning of NTK's

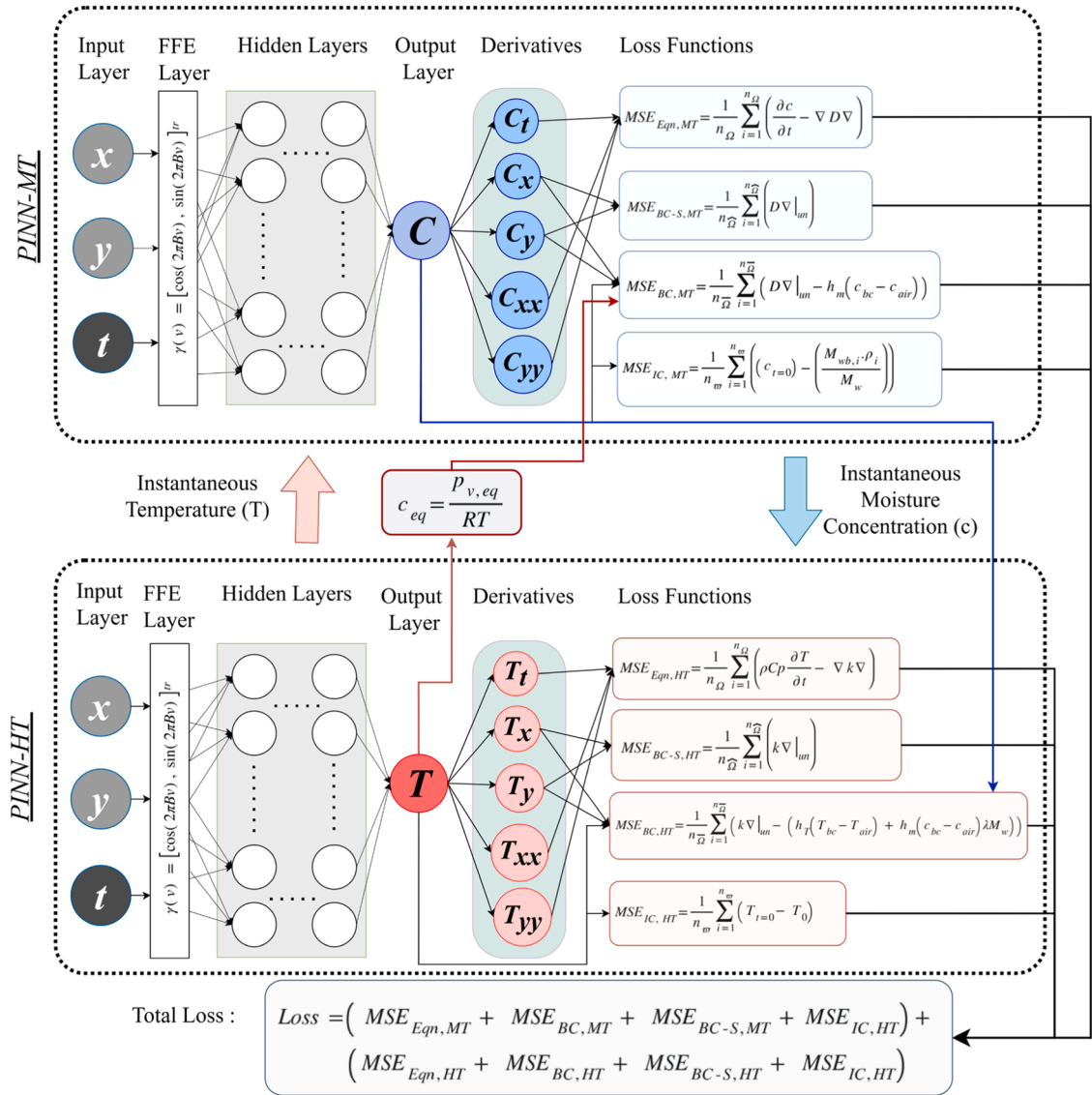


Fig. 2. Schematic diagram of the coupled PINN-MT and PINN-HT architecture, illustrating spatiotemporal inputs, outputs for moisture concentration and temperature, and the relevant residual loss functions.

spectrum by modifying the frequency vector during the training process. By adjusting the random Fourier feature mapping from Gaussian distribution and corresponding kernel width, both convergence rates and prediction accuracy can be significantly improved. Additionally, this development positively impacts the generalization behaviour of the network.

The FFE presents a remarkably straightforward approach to input transformation. At its core, it takes an input point v (such as x, y, t in this study) and transforms it using a randomly initialized matrix B . This matrix B is constructed by sampling values from a Gaussian distribution with mean zero and variance σ^2 . The transformation then applies sine and cosine functions shown below [53].

$$\gamma(v) = [\cos(2\pi Bv), \sin(2\pi Bv)]^T \quad (22)$$

Moreover, the FFE layer fundamentally addresses the loss imbalance issue that typically requires manual weighting in standard PINNs. By mapping the physical coordinates (x, y, t) to a selected high-dimensional frequency space (256 encoded dimensions with $\sigma=0.01$ in this application), FFE enables the network to capture multi-scale features and high-frequency components more effectively from the outset, leading to more balanced gradient flow across residuals from equations during

training. This automatic balancing eliminates the need for explicit loss weighting or adaptive coefficients. A sensitivity analysis on loss weights was therefore deemed unnecessary for this FFE-PINN framework, consistent with recent studies demonstrating improved training stability through Fourier embeddings. Interested readers may refer to [53] for further details on the incorporation of random Fourier feature embedding into MLPs.

Thanks to these development, this is the first time to develop PINN framework to solve coupled heat and mass transfer without weight initialisation through transfer learning [55] and customisation of loss weights [56] in food drying context while reaching the accuracy achieved in this paper. For the experiments in this study, both the PINN-MT and PINN-HT models use the Gaussian Fourier Feature Embedding layer with 256 encoded input dimensions and a standard deviation of 0.01, applied to the physical coordinates. Each embedded input is then processed by a fully connected network consisting of 4 hidden layers with 128 neurons per layer and sine activation, followed by a single output neuron for the moisture concentration in PINN-MT network and temperature in PINN-HT network. The domain consists of 441 spatial points with 60 timesteps, with each time step corresponding to 1 min. The learning rate was set to $1e^{-4}$, without the use of any scheduler. All

Algorithm

Coupled FFE-PINN training loop with real time updates.

```

for epoch = 1 to Epoch # Repeat
  # Forward pass: Evaluate both networks at all collocation points (x,y,t)
  Calculate C, C_x, C_y, C_xx, C_yy, C_t for PINN-MT(x,y,t)
  Calculate T, T_x, T_y, T_xx, T_yy, T_t for PINN-HT(x,y,t)
  # Coupling Step 1:
  Compute  $P_{v,eq}$  using current temperature (T) using Eq. (11)
  Update moisture concentration C.
  # Coupling Step 2: Compute  $C_p(C)$  using current C_pred
  Calculate wet basis moisture content ( $M_{wb}$ )
  Calculate specific heat ( $c_p$ ) and thermal conductivity ( $k$ ) based on updated wet basis moisture content ( $M_{wb}$ )
  # Compute mean squared residuals for PINN-MT
  MSEEqn,MT: Residual from mass transfer governing equation – Eqn (14)
  MSCBC,MT: Residual from mass transfer boundary condition – Eqn (15)
  MSCBC-s,MT: Residual for symmetric line – Eqn (16)
  MSCIC,MT: Residual from mass transfer initial condition – Eq. (17)
  # Compute mean squared residuals for PINN-HT
  MSEEqn,HT: Residual from heat transfer governing equation – Eq. (18)
  MSCBC,HT: Residual from heat transfer boundary condition – Eq. (19)
  MSCBC-s,HT: Residual for symmetric line – Eq. (20)
  MSCIC,HT: Residual from heat transfer initial condition – Eq. (21)
  # Compute losses
  MT_loss = MSEEqn,MT + MSCBC,MT + MSCBC-s,MT + MSCIC,MT
  HT_loss = MSEEqn,HT + MSCBC,HT + MSCBC-s,HT + MSCIC,HT
  Total loss = MT_loss + HT_loss
  Update NN parameters through backpropagation
  Epoch = Epoch + 1
end

```

numerical experiments were performed on an Intel(R) Core(TM) i5–12, 500 CPU with 32 GB of RAM, with an RTX 3060 GPU, with results compared against experimental findings [57] and against COMSOL FEA [58] model for validation.

4. Results

This section discusses the performance of the developed PINN framework, firstly highlighting the effects of the FFE-layer integration into NN and secondly investigating the heat and mass transfer characteristics. The investigation focuses on apple as the drying material, and the relevant specific property parameters are provided in Table 1.

Homogeneous product properties and constant process conditions are assumed in this investigation by using constant drying temperature (T), diffusion coefficient (D), heat transfer coefficient (h_r) and mass transfer coefficient (h_m), in order to minimise additional complexities and focus on the coupled FFE-PINN formulation. The influence of product heterogeneity and variable drying conditions on drying behaviour has been investigated by authors previously using transfer learning in PINN-MT mass transfer analysis computational model [24] as well as PINN-S shrinkage analysis computational model [44].

Table 1

Physical property parameters for coupling the PINN-MT and PINN-HT models.

Parameter	Value	Reference
Diffusion coefficient $[D]^*$	4.19×10^{-10} (m ² /s)	[8]
Initial tissue dry basis mass $[M_{dbi}]$	6.40 (kg/kg _{dm})	[21]
Intracellular water percentage $[M_{ICWI}]$	87.8 %	[59]
Molar mass of water $[M_w]$	0.018016 (kg/mol)	[2]
Universal gas constant $[R]$	8.314 (J/(mol.K))	[60]
Density of apple $[\rho]$	837 (kg/m ³)	[61]
Moisture concentration in air $[C_{air}]$	0.654 (mol/m ³)	[34]
Partial vapor pressure $[P_{v,air}]$	2000 (Pa)	[21]
Heat transfer coefficient $[h_r]$	15.21 (W/m ² /K)	[21]
Mass transfer coefficient $[h_m]$	10 (m/s)	Set*
Drying temperature $[T_{air}]$	333.15 (K)	Set*
Initial temperature $[T_0]$	293.15 (K)	Set*
Latent heat of evaporation $[\lambda]$	2358,600 (J/kg)	[21]

*These values were selected to investigate the heat and mass transfer variations during the drying of 20 mm × 20 mm apple tissue over a 60-minute period.

4.1. Effects of fourier feature-embedding (FFE) in PINN

The PINN framework was enhanced with a FFE layer to handle complex coupled heat and mass transfer physics in 2D tissue drying over 60 min as discussed in Section 3.4. The computational accuracy of PINN relies heavily on derivatives computed through AD in neural networks. FFE layers specifically improve the accuracy of high-frequency function derivatives within low-dimensional domains. To demonstrate this enhancement, we tested an FFE-enabled neural network with cos activation function by learning a SEM image from the small tissue experimental study [62] (data function of the original image can be defined as $f(x)$) along with its first derivative ($\nabla f(x)$) and second derivative ($\nabla \nabla f(x)$), comparing results against standard numerical computations as in Fig. 3. All network settings were kept same except the FFE layer where the NN is set to train with 256 input dimensions and a standard deviation of 1.

The results shown in Fig. 3 provides a comprehensive comparison between numerical benchmarks and neural network predictions. The left column establishes the baseline, featuring the original SEM image (top) and its numerical first- and second-order derivatives (middle and bottom). When comparing prediction capabilities, the standard NN without FFE layer and transfer learning weight initialization (middle column) demonstrates significant limitations even after 10,000 training iterations. This is evidenced by its poor reproduction of the original image and severely limited detail capture in both first- and second-order derivatives. In contrast, the FFE-NN implementation (right column) shows remarkable improvement in prediction accuracy across all images. Without any weight initialization, the FFE-NN successfully captures intricate details that closely match both the original image and its numerical derivatives. This difference in performance clearly demonstrates the FFE layer's effectiveness within the NN learning capabilities towards first and second derivatives along with automatic differentiation. The FFE layer's ability to enhance learning capacity and achieve accurate predictions without additional initialization requirements underscores its novel contribution to the methodology of this paper.

4.2. Validation against experiments

Next, the performance of PINN-MT for tissue domain is evaluated

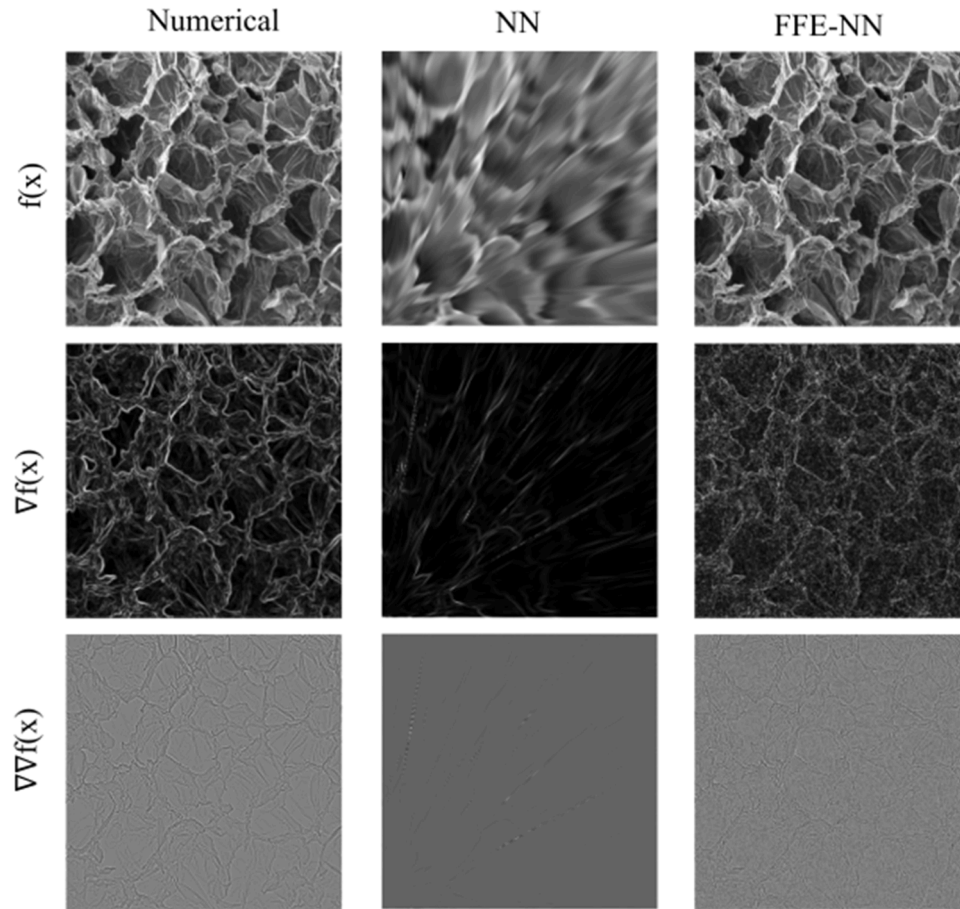


Fig. 3. Comparison of computing for an original SEM image, its first- and second-order derivatives through numerical differentiation and automatic differentiation without and with FFE layer in NN.

and validated against experimental results published in [57]. This experimental study focused on small tissues with 70 °C drying temperature and hence drying time is 42 mins (<60 mins) which is conveniently achievable within the capacity of the developed PINN framework in this study. Following the tissue sizes in that experimental study, the full tissue size is considered as 10 mm x 10 mm and a quarter model of 5 mm x 5 mm domain including symmetric boundary conditions at left and bottom boundaries as described in Fig. 1, with $h_m = 15$ m/s for the results in this section. The mean dry bases moisture content variation from PINN framework is plotted and compared against experimental results as seen in Fig. 4.

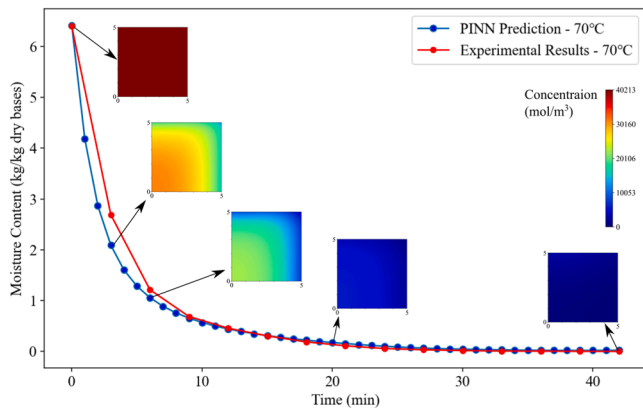


Fig. 4. Mean moisture content (kg/kg dry bases) against the drying time (min) comparison against experiment in [57].

As depicted in Fig. 4, the PINN framework's moisture content predictions (blue line) align closely with the experimental results (red line) at 70 °C. The PINN prediction follows drying rate variations accurately throughout the total drying duration of 42 min. The embedded contour plots show the PINN's spatial moisture distribution at different time intervals ($t = 0, 3, 6, 20$, and 42 min), transitioning from high moisture content (dark red) to low moisture content (dark blue). These distribution plots effectively visualize how moisture content decreases both over time and spatially across the sample, with the colour scale ranging from 0 to 40,213 mol/m³ as seen in the scale.

4.3. Validation against computational models

In order to validate the PINN-MT and PINN-HT coupled PINN framework along with temperature variations within a drying duration of 60 min, the results are compared against the FEA model developed in [21] with this study settings. The analyses were based on the drying kinetics of macroscopic food tissues with the size of 20 mm × 20 mm. When this size of an apple tissue is dried in a convective dryer at a medium drying temperature of 60 °C (333.15 K) for 60 min, the variations in heat and mass transfer are analysed, focusing on moisture concentration and temperature distribution.

4.3.1. Concentration profiles

Moisture content in dried food materials is a critical parameter that represents the amount of moisture relative to the mass of the completely dry material or dry matter, typically expressed in kg/kg on a dry bases. This parameter is essential for analysing the drying process of plant-based foods and plays a key role in maintaining the desired texture,

taste, and overall quality of the food. In this study, the moisture prediction capabilities of the PINN framework were compared against the predictions from a comparable FEA-based numerical model, as shown in Fig. 5.

As shown in Fig. 5, the moisture content evolution during the drying process exhibits two distinct phases: an initial rapid decrease followed by a more gradual drying phase. The sharp initial decrease in moisture content (from approximately 6.4 to 4 kg/kg dry basis within the first 5 min) can be due to the higher mass transfer coefficients set at the boundaries, creating a steep concentration gradient in both FEA and PINN results. Subsequently, the drying rate transitions to a more gradual decline, with both PINN predictions and FEA simulations showing similar trends. The PINN framework demonstrates good agreement with the FEA results throughout the drying period, though a small consistent offset is observed. This minor deviation between PINN predictions (red dashed line) and FEA simulations (blue solid line) remains relatively constant throughout the drying process, particularly in the gradual drying phase, suggesting systematic differences in how each method handles boundary conditions and spatial derivative calculations. Overall, the graph shows that PINN framework effectively captures both the dramatic initial moisture loss and the subsequent slower drying dynamics, showcasing the PINN's drying kinetics prediction capabilities over the full 60-minute period. Further comparisons have been carried out based on the spatial moisture concentration variations for selected time steps as shown in Fig. 6.

As seen in Fig. 6, the contour plots compare the spatiotemporal evolution of moisture concentration between FEA simulations (Fig. 6-a) and PINN predictions (Fig. 6b) at four different time intervals (1, 20, 40, and 60 min), with their percentage difference shown in Fig. 6c. At $t = 1$ min, both methods show uniform high concentration, indicating the initial saturated state. As drying progresses ($t = 20, 40$, and 60 min), a distinctive moisture gradient develops from the boundaries inward, characterized by a sharp transition near the edges while maintaining higher concentration in the core region. The %Error in Fig. 6c highlights that the evident discrepancies occur near the boundaries, with maximum deviations below 10 %. Such differences near boundaries can be due to the difference between FEA's mesh-based numerical differentiation and PINN's mesh-free automatic differentiation approaches in handling steep concentration gradients. Despite these localised differences, the overall moisture distribution patterns show strong agreement between both methods, with the majority of the domain showing errors below 2 %, demonstrating PINN framework's robust capability in capturing complex moisture transport phenomena. The spatial

differentiation effects are further illustrated in the 3D contour plots presented in Fig. 7 below.

Fig. 7 provides a more detailed 3D visualization of the moisture concentration evolution within the tissue domain. The surface plots reveal the complete spatial distribution patterns. In the FEA results (Fig. 7a) the contour surfaces demonstrate remarkably smooth and uniform moisture gradients, particularly evident in the initial condition ($t = 1$ min) where the surface appears nearly planar. The PINN predictions (Fig. 7b) capture the same overall trends, though with slight variations. As drying progresses, both methods show a clear development of steep moisture gradients near the boundaries, visualised by the sharp transitions from red (high concentration) to blue (low concentration) regions. The PINN framework's predictions increasingly align with the FEA results in later stages, demonstrating its capability to stabilize and accurately capture the physical behaviour of moisture transport. These 3D visualisations provide additional insight into the boundary effects previously observed in Fig. 4, confirming that while minor discrepancies exist in the predictive approaches, they don't significantly impact the overall drying behaviour prediction.

4.3.2. Temperature profiles

This section presents the temperature variation predictions from the coupled heat transfer computational model, integrated in the PINN framework. The average temperature at each drying time was calculated and plotted for comparison and validation of the PINN-HT results against FEA results. As discussed in Section 2, the thermal conductivity and specific heat of the material are functions of moisture content. Consequently, the variation trends observed in moisture content are similarly reflected in the average temperature profiles.

As per Fig. 8, the tissue experiences a characteristic heating curve as it is exposed to drying air at 60 °C from initial temperature of 20 °C. The temperature profile shows two distinct phases: an initial rapid heating period (0–20 min) where the temperature rises sharply from 20 °C to approximately 42 °C, followed by a more gradual increase toward the equilibrium temperature. Both PINN predictions (red dashed line) and FEA simulations (blue solid line) demonstrate excellent agreement throughout the drying period, with only minor deviations during the rapid heating phase. These small discrepancies can be attributed to the different approaches in handling the coupled heat and mass transfer calculations between the FEA (calculates in each timestep) and PINN frameworks (calculates in each training iteration). Overall, this figure indicates the robustness of the PINN framework in capturing the heat transfer characteristics during the drying process in a comparable

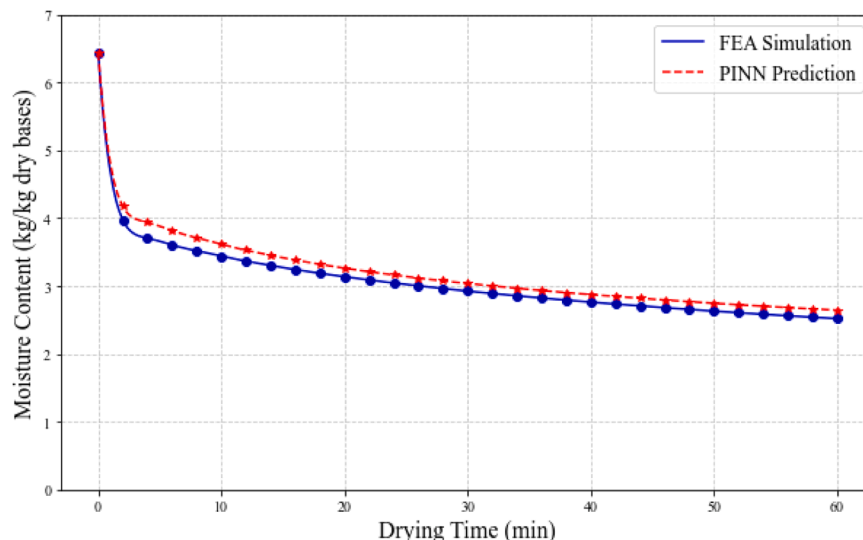


Fig. 5. Mean moisture content (kg/kg dry bases) against the drying time (min) comparison against FEA.

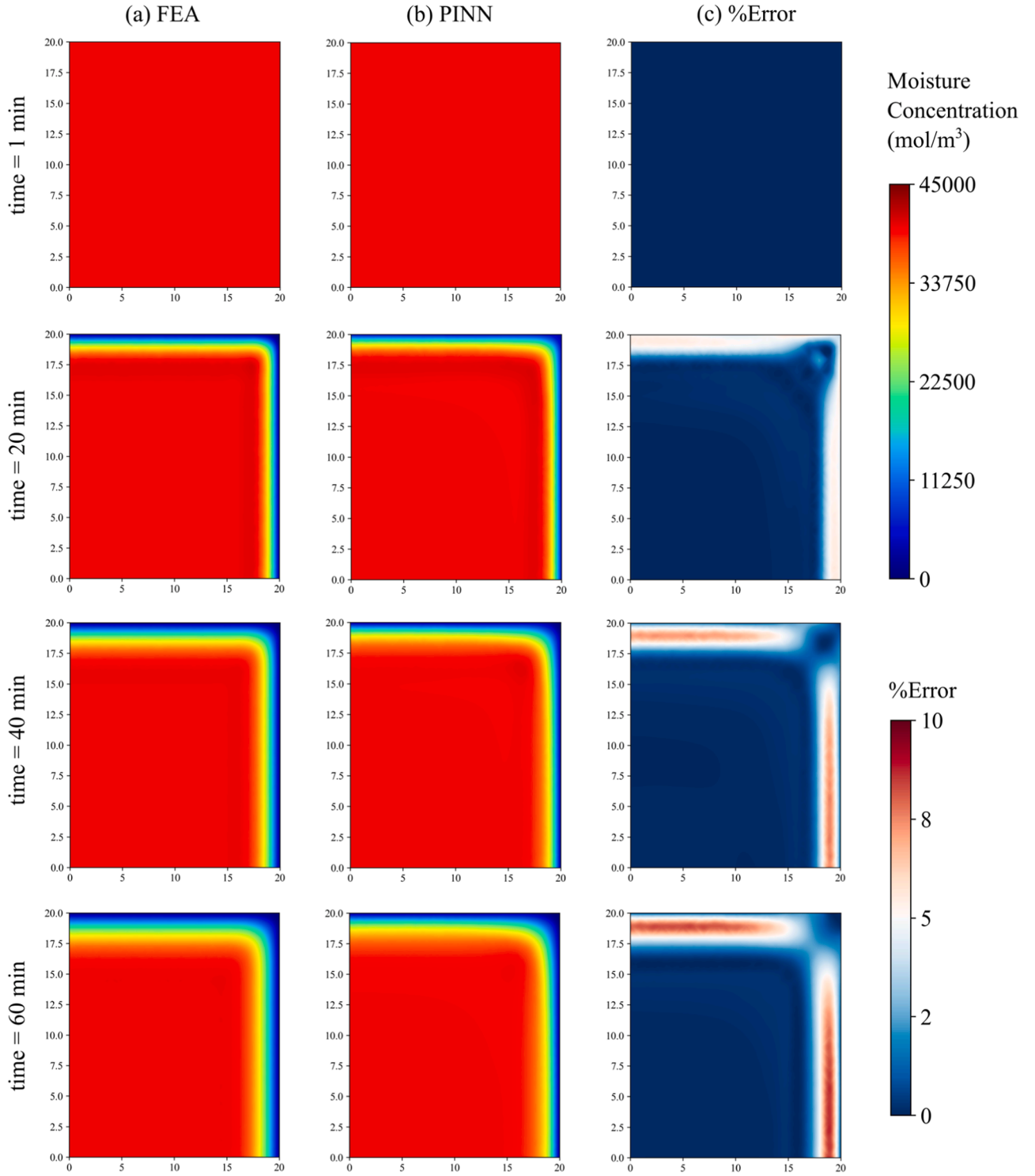


Fig. 6. Spatial moisture concentration (mol/m^3) variations of the drying tissue. (a) Predictions from the FEA model, (b) Predictions from PINN framework, and (c) deviation as the %Error.

manner to established FEA numerical approaches. Detailed spatial temperature distributions at selected time intervals are further examined in Fig. 9.

Fig. 9 provides a detailed visualisation of the spatiotemporal temperature evolution through 2D contour plots, comparing FEA simulations (Fig. 9a), PINN predictions (Fig. 9b), and their percentage differences (Fig. 9c) at the same four selected timesteps during the drying process. The temperature field ranges from 293 K to 333 K (20°C to 60°C), represented by a blue-to-red colour gradient. At $t = 1$ min,

while FEA shows a uniform low-temperature distribution, PINN predicts slight temperature elevations near the boundaries, resulting in the maximum deviation of 4.51 % in overall study. This early-stage difference suggests that the proposed PINN-based heat and mass transfer predictive model captures the initial heat diffusion dynamics more sensitively than FEA. As drying progresses ($t = 20, 40$ min), both methods demonstrate the development of clear temperature gradients from the boundaries inward, though PINN shows slightly different gradient patterns with %Errors typically ranging from 1–2 %. By $t = 60$

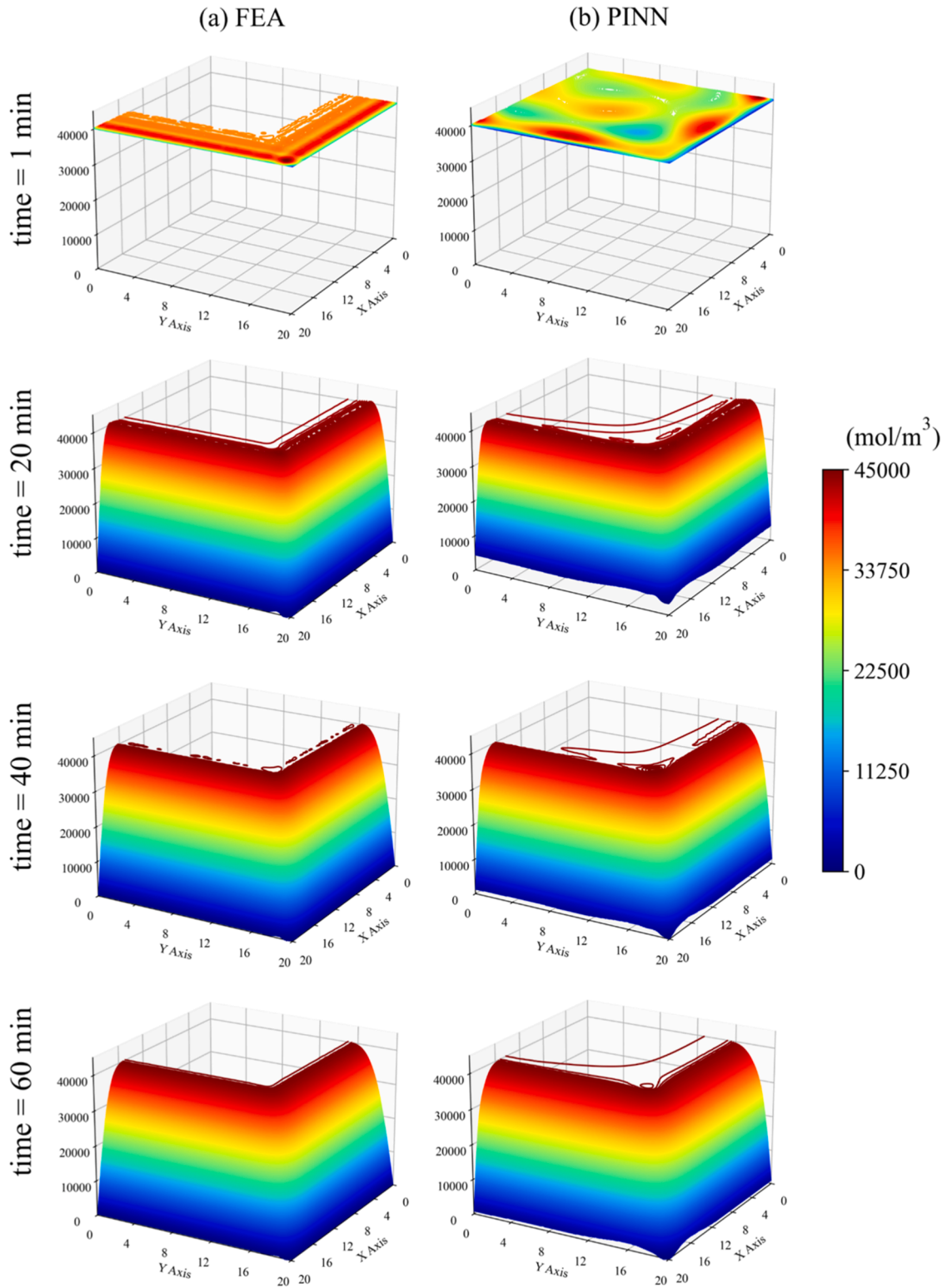


Fig. 7. 3D contour visualisation of spatial moisture concentration variations (mol/m^3) within the tissue. (a) Predictions from the FEA model and (b) Prediction from the PINN framework.

min, both methods predict similar overall temperature distributions approaching the outside drying temperature, though PINN maintains slightly less uniform patterns in high-gradient regions. These differences, visualised in the error maps (Fig. 9c), highlighting the different approaches to calculate the spatial gradients between FEA and PINN frameworks. For a more comprehensive understanding of these thermal

patterns, three-dimensional temperature distributions are presented in Fig. 10.

Fig. 10 provides clear visualization through 3D surface plots, offering deeper insights into the thermal evolution predicted by both methods. As the drying progresses, both methods capture the fundamental physics of heat transfer, showing the progressive warming from

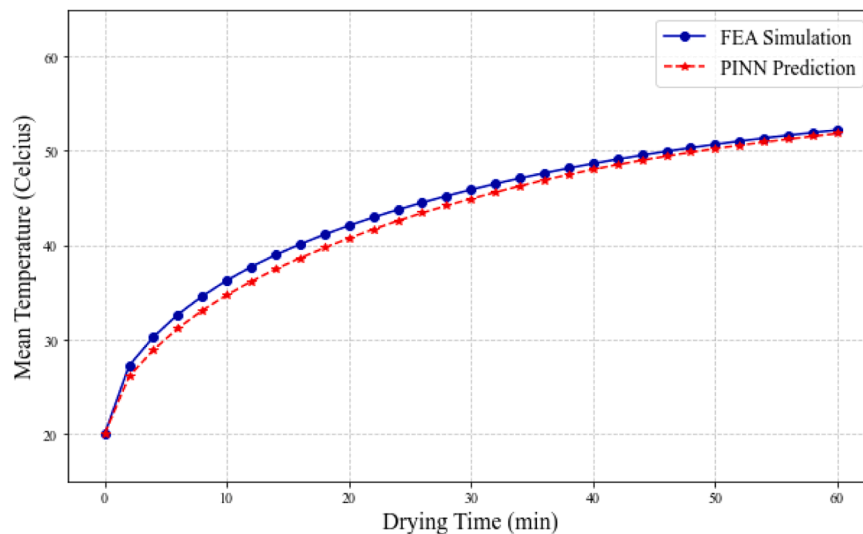


Fig. 8. Mean temperature (°C) against the drying time (min) comparison against FEA.

boundaries inward. However, they slightly differ in their representation of temperature gradients. These gradients, visualized as surface slopes in the 3D plots, show how each method handles thermal diffusivity within the plant tissue. These differences in gradient calculations, while minimal in terms of absolute temperature predictions, highlight the distinct derivation approaches utilised in both techniques.

5. Discussions

The proposed PINN framework demonstrates robust capabilities in predicting both moisture concentration and temperature distributions during the drying process of plant tissues, despite being a relatively new deep learning-based computational technique. To the authors' best knowledge, this work represents the first development of a mutually coupled heat and mass transfer PINN-based computational model investigating plant tissue drying for extended periods (60 min) without labelled data. While Manavi, Fattahi and Becker [40] recently developed a heat and mass transfer surrogate PINN model, their work was limited to 5 min of drying time for a 1D domain and did not consider mutual effects between temperature and concentration. Our achievement in developing a PINN framework for predicting coupled heat and mass transfer effects in practical applications underscores the significant potential of this computational framework.

Comparative analyses against traditional FEA methods reveal minor discrepancies, particularly in spatial gradient calculations near boundaries (maximum errors of 8.73 % for moisture and 4.51 % for temperature). These differences primarily arise from the distinct approaches to differentiation, automatic in PINNs versus numerical in FEA, rather than from inherent limitations of the PINN approach (refer to Appendix A). The PINN framework provides naturally continuous solutions without mesh-induced domains and successfully captures physical phenomena in both mass and heat transfer processes, establishing itself as a promising alternative to traditional numerical computational methods. However, computational efficiency remains a challenge for the PINN-based model, requiring approximately 21.23 min for complete training without weight initialisation compared to COMSOL FEA's 5-second simulation time. This training time highly depend on the collocation points as discussed in Appendix B, while the current 2D model uses 441 spatial points in 60-time steps which is over 26,000 collocation points, which however fits comfortably within 12 GB GPU memory. While FFE maintains the same network architecture regardless of spatial dimensionality (fixed 256 encoded input dimension), scalability to larger time steps or to 3D would increase the number of collocation points, potentially requiring more GPU memory and training iterations proportional

to the increased dataset size.

Yet, it is also worth to note that PINNs are capable of training across a wide range of parameter variations and once trained, it is capable of making predictions within a fraction of a second, while an FEA-based model requires re-computation with each parameter adjustment [24], highlighting the differences in the two computing paradigms. It also needs to highlight that FEA has benefited from decades of optimization, including developments such as weak form formulation for enhanced numerical stability and error control. In contrast, PINNs are relatively new to physics-based problem-solving and are typically implemented in basic Python code using neural network architectures borrowed from data science. While still evolving, PINNs offer significant advantages particularly for problems with highly nonlinear and heterogeneous parameter variations such as in biological systems [34,35,63,64]. Their inherent capabilities include; multiphysics and data coupling, surrogate modelling [39,40], and boundary integration capabilities [34,65]. Also PINNs are meshless, eliminating mesh generation and refinement complexities in moving boundaries [66] and capable of naturally handling inverse problems with no additional steps requirements [67]. On top of all these, PINNs are capable of seamless incorporation of experimental data into the predictive frameworks allowing for the blending of limited revealed physics with data-driven learning, which can lead to revelation of unknown physics and subsequent parameters. Moreover, they are capable of combining deep learning computer vision and other capabilities, providing sophisticated insights for problems where the underlying physics are not yet fully known [31]. However, significant improvements in NN parameters similar to FFE in this study, are still required to enhance predictions of heat and mass transfer variations for complete real drying cycles with significant nonlinearities and heterogeneity due to varying cell sizes, porosity gradients, and intracellular/extracellular water partitioning, as demonstrated in [21]. Given these capabilities, PINNs represent a potential paradigm shift, enabling broader applications in food processing and other soft matter biological applications, particularly where multiple physical phenomena interact simultaneously with transient, nonlinear and heterogeneous effects impacting both processes and products.

6. Conclusion

This study proposes a novel framework, Fourier Feature Embedded Physics-Informed Neural Networks (FFE-PINN), for the analysis of complex heat and mass transfer phenomena associated with drying of plant tissues. The proposed framework effectively integrates two physics-informed neural network models, PINN-MT for mass transfer

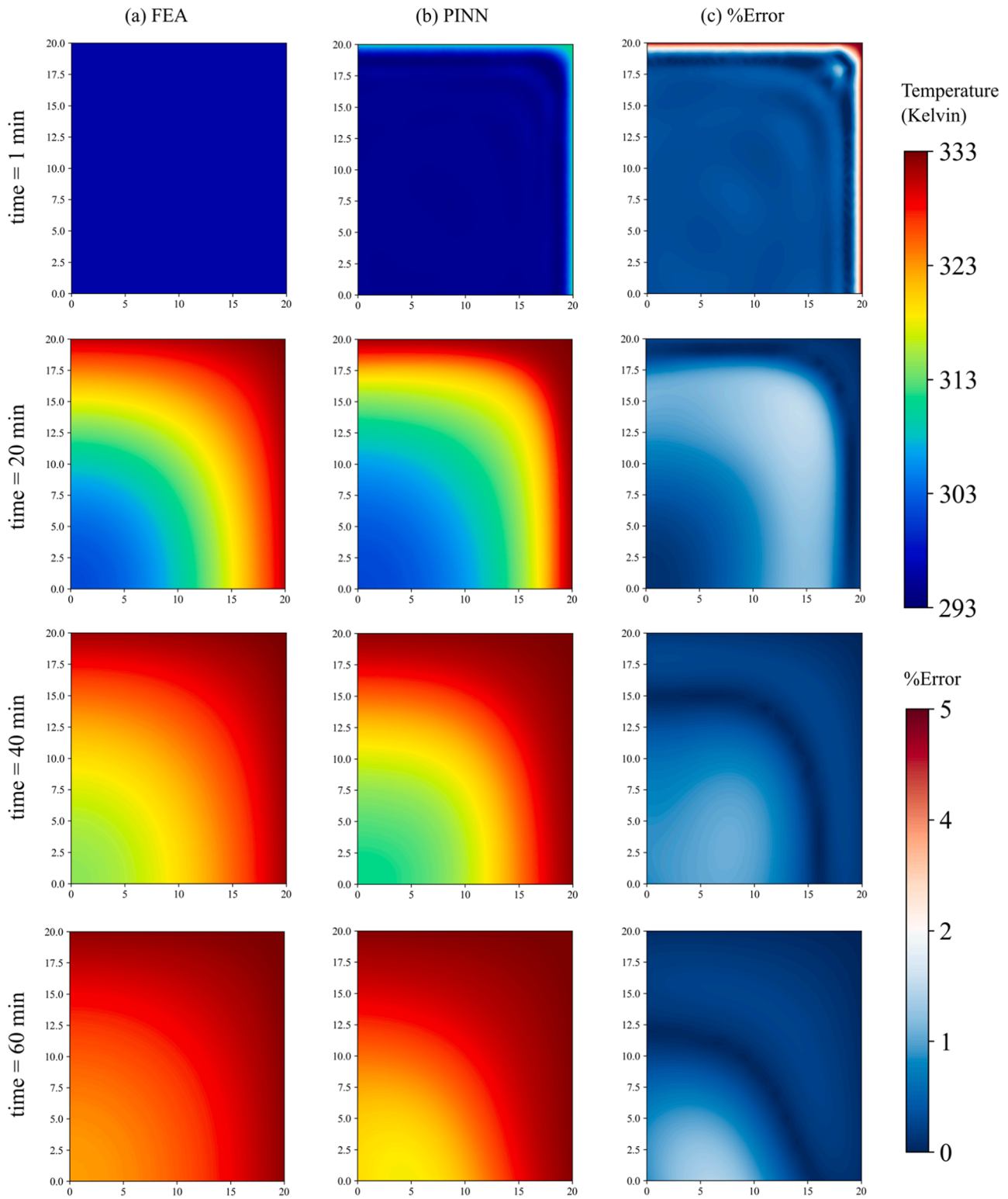


Fig. 9. Spatial temperature distribution (K) within the tissue. (a) Predictions from the FEA model, (b) Predictions from the proposed PINN framework, and (c) deviation as the %Error.

and PINN—HT for heat transfer, demonstrating strong predictive capabilities in capturing spatiotemporal variations of moisture content and temperature distributions, while accounting for their mutual interactions. A key contribution of this study is the integration of a Fourier Feature Embedding (FFE) layer within the PINN framework to investigate coupled heat and mass transfer during drying, substantially enhancing the predictive performance, enabling accurate predictions

throughout a realistic 60-minute drying period, being the first of its kind to achieve such progress without requiring additional support from weight initialization or transfer learning techniques.

A comprehensive comparative analysis against experimental data and traditional FEA findings validates the effectiveness of the proposed PINN framework, demonstrating strong agreement with maximum deviations of 8.73 % for moisture concentration and 4.51 % for

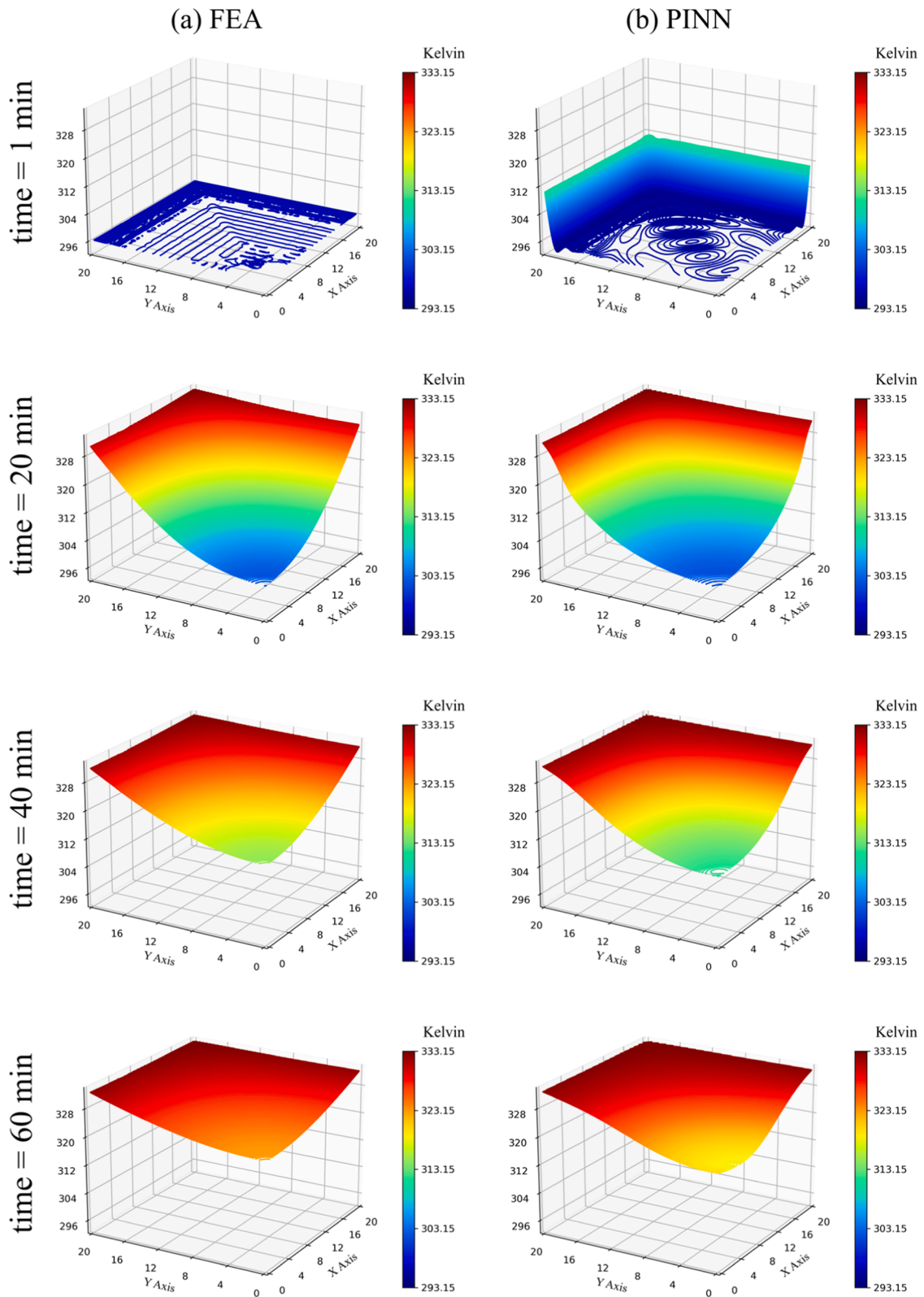


Fig. 10. 3D contour visualization of spatial temperature distribution (K). (a) Predictions from the FEA model and (b) Predictions from the PINN framework.

temperature predictions across the entire spatiotemporal domain. As discussed above, even these small discrepancies, localized near boundaries with Neumann conditions, arise due to the fundamental differences between PINN's automatic differentiation and FEA's numerical differentiation techniques rather than due to any limitations of the proposed PINN framework. The proposed FFE-PINN framework's successful

handling of coupled nonlinear physics, particularly in terms of managing temperature-dependent material properties and complex boundary conditions concurrently, marks a significant advancement in computational modelling of food drying processes and heat and mass transfer processes through PINNs.

There is potential for further research to focus on expanding the

proposed framework's capabilities to incorporate additional physical phenomena, such as mechanical deformation during drying, multiscale modelling aspects related to cellular structure changes at different scales including 3D real sample scale, and sensitivity against drying environment changes. Capabilities of PINN's and FFE could be further leveraged through integration of individual-problem-specific experimental data and optimization algorithms to improve practical applicability in specific industrial food processing operations as well as in other analogical applications related to heat and mass transfer.

CRedit authorship contribution statement

C.P. Batuwatta-Gamage: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **H. Jeong:** Writing – review & editing, Software. **ZG Welsh:** Writing – review & editing,

Resources. **M.A. Karim:** Writing – review & editing. **H.C.P. Karunasena:** Writing – review & editing. **C.M. Rathnayaka:** Writing – review & editing. **Y.T. Gu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Numerical vs Automatic Differentiation

This is to show the derivative difference between automatic differentiation against numerical FEA derivation.

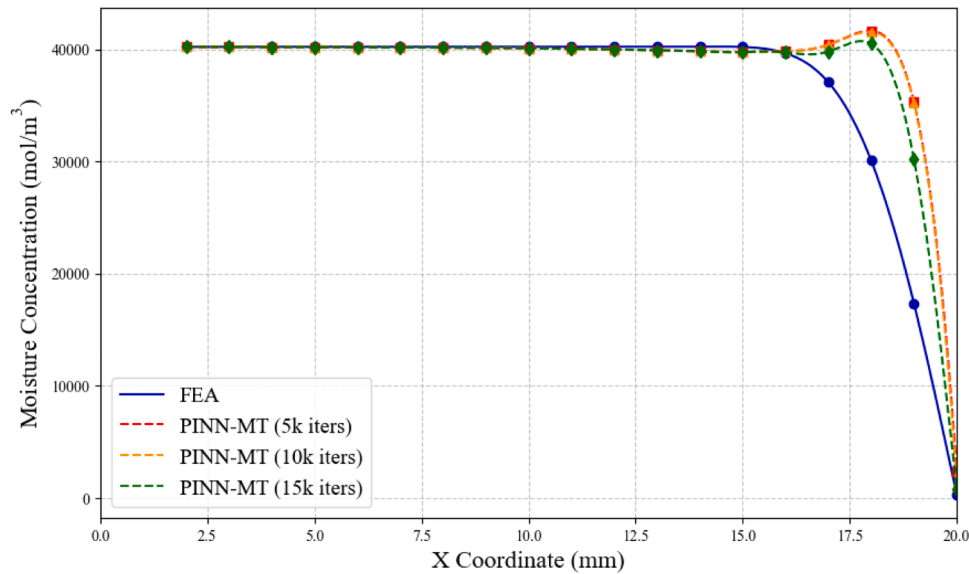


Fig. A1. Moisture concentration variation along X-axis along the mid height boundary ($y = 0$) at $t = 60$ min comparing FEA and smoothness against training iterations in PINN.

Fig. A1 provides a one-dimensional cut of the moisture field along the middle boundary (when $y = 0$) when $t = 60$ min, clearly illustrating that the largest discrepancies between FEA and PINN-MT occur in the high-gradient (dc/dx) region close to the exposed surface. This is aligned with the %Error variation shown in the **Fig. 6(c)**. The discrepancy in this high-gradient region varies slightly and reduces as the PINN is trained for more epochs (compared changes from 5 000 to 15 000 iterations), although this improvement comes at the cost of a significantly increased training time. Because the gradient field is symmetric, the middle vertical boundary at $x = 0$ exhibits a similar trend for gradient variations (dc/dy). The same effect is also evident in the time-domain derivatives in the high-gradient region during the early stage of drying (approximately 1–2 min), as seen in **Fig. 5**.

In the PINN solution, spatial derivatives are obtained by automatic differentiation of a smooth neural network representation of $c(\mathbf{x}, \mathbf{y}, t)$, whereas the FEA gradients are computed from nodal values using mesh-dependent numerical differentiation. Consequently, the FEA solution is more sensitive to local mesh resolution in regions with sharp gradients, while the PINN field remains smooth but can slightly diffuse these gradients, leading to the observed deviations near the boundary despite close agreement in the domain interior.

Appendix B

Training Time Comparison against Domain Scales

This additional investigation carried out to analyse training time variations against domain scales.

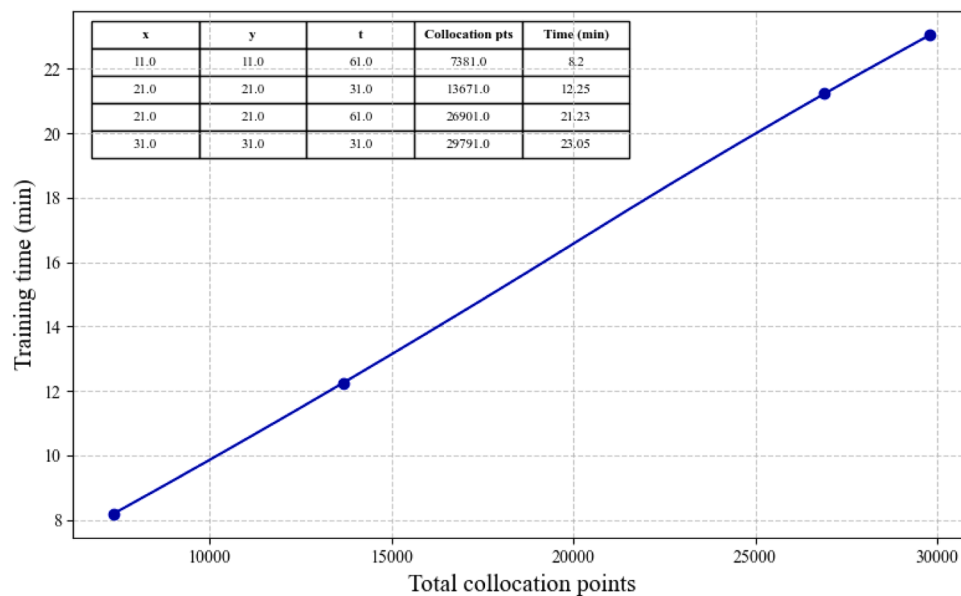


Fig. B1. Training time variation against Domain Scales.

As training time mainly depends on the total number of collocation points used to represent the domain, rather than on the physical size of the domain itself, the same spatial extent (20 mm × 20 mm) was retained while varying the spatial and temporal resolutions, as summarised in Fig. B1. In each case, the neural network architecture and optimiser settings were kept identical; therefore, the observed increase in training time reflects the higher collocation-point density (from 7381 to 29 791 points) for 10,000 training iterations rather than any change in model complexity. This supports the statement in Section 5 that, for a fixed architecture, FFE-PINN training time scales approximately linearly with the number of collocation points.

Data availability

Data will be made available on request.

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